

Visualization Methods for Manual Wheelchair Training: Impact on Communication Between Coaches and Users

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Abstract

This study evaluates the effectiveness of a proposed visualization method for enhancing communication regarding skill improvement between wheelchair users and coaches. Two measurements were conducted to observe changes in communication dynamics and skill development. The study involved recording and analyzing a coach's evaluation of a user's 40-m sprint utilizing video, pose estimation, and sensor data graphs. The interactions between the coach and the user were subsequently analyzed to elucidate how communication evolved and to identify any improvements in the skills of users. The findings suggest that integrating these visualization tools may enhance discussions on skill enhancement and contribute to more effective training outcomes.

CCS Concepts

• **Human-centered computing** → **Information visualization**.

Keywords

Manual wheelchair operation training, data visualization, evaluation of manual wheelchair users

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1 Introduction

Mastering wheelchair operation techniques is vital for improving physical potential and mobility safety for all users, including athletes. However, traditional methods like video analysis and stopwatch timing lack the quantitative data needed to fully support communication on skill development. This study addresses this issue by offering tools that visualize the movement of both wheelchairs and users, enhancing communication between coaches and users during training.

2 Methods

2.1 Kinematic tracking of wheelchair operation training

Recent advancements in sensor technology and image analysis have enhanced support for physical activities[Cotton 2020; Uno et al. 2023]. In wheelchair training, visualizing posture and movement efficiency is essential for improving communication between coaches and users. Accurate sensing of movements and postures is

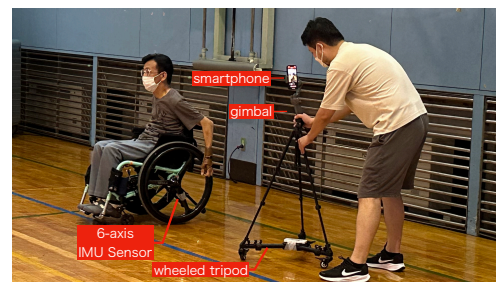


Figure 1: Measurement environment (6-axis IMU Sensor: measures wheelchair movement, smartphone: records user movement)

key. This study utilizes the ORPHE ANALYTICS system, which integrates IMU sensors with pose estimation from smartphone-recorded videos, providing customizable data visualization to enhance communication.

As shown in Figure 1, 6-axis IMU sensors (sampling rate: 200 Hz) were attached to both large wheels of the wheelchair. A smartphone with a gimbal, mounted on a wheeled tripod, recorded the wheelchair's 40-m sprint from approximately 1 m away, moving alongside the user.

2.2 Data visualization

In collaboration with a wheelchair coach, we prioritized visualizing three elements (Figure 2)—video, pose estimation from video, and sensor data graphs to assess their effectiveness in wheelchair operation training:

- Video: A familiar instructional tool for both coaches and users.
- Pose estimation from video: Colored bars on the body help verify hand trajectories and upper body posture, improving communication.
- Sensor data graphs: Wheel angular velocity measurements show speed changes during each push, aiding in high-quality communication.

While joint coordinates from pose estimation were considered, the accuracy of these estimates from parallel-running video footage is uncertain, requiring further investigation.

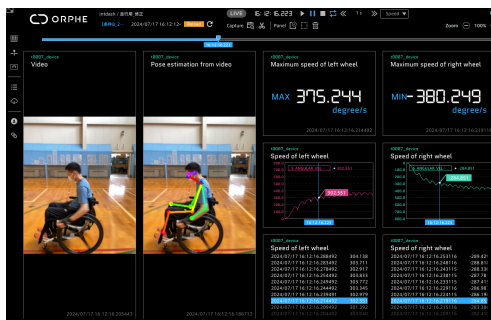


Figure 2: Data visualization (left: video, middle: pose estimation from video, right: sensor data graphs)

2.3 Evaluation experiment

This study evaluated the wheelchair skills of 12 users (10 males, 2 females) through periodic training every three months for those who had undergone prior training.

After the 40-m sprint assessment, the coach used an 11-inch iPad Pro with the ORPHE ANALYTICS Viewer (Figure 3) to review videos, posture analysis, and sensor data, providing feedback on movement techniques and improvements. Two measurements were taken, separated by over three months, to assess if the visualization method enhanced communication about skill improvement.

Additionally, the coach's feedback during the assessment was recorded and analyzed, incorporating video, pose estimation, and sensor data graphs. A conversation analysis was then performed to examine the communication between the coach and the user.



Figure 3: Assessment of wheelchair operation techniques

3 Result and discussion

Of the twelve wheelchair users, one was hospitalized and could not participate in the second measurement, and sensor data for another were missing in the first measurement. The analysis included data from 10 users (eight males, two females).

Observation of Wheelchair Coach

Video analysis was used to evaluate nine users' propelling techniques, hand trajectory, and posture. Pose estimation was performed for three users to assess their posture and hand movements. Sensor data graphs were analyzed for all ten users, focusing on maximum speed, stroke count changes, and speed variations.

Observation of Wheelchair Users

Two users discussed changes with the coach, and one noted asymmetry in the wheelchair wheels' maximum speed. The coach used video and sensor data to assess the 40-meter sprint, leading to new evaluation methods like explaining speed changes, identifying technique factors, analyzing wheel discrepancies, and developing strategies. Wheelchair users also discussed technique improvements with the coach using these visualizations.

4 Conclusion

This study introduced a method to visualize wheelchair operation techniques during a 40-m sprint, enhancing communication between coaches and users by integrating sensor data graphs with traditional video analysis. While pose estimation from videos was not used, future research should explore its application to further improve training methods.

Acknowledgments

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References

- R. James Cotton. 2020. Kinematic Tracking of Rehabilitation Patients With Markerless Pose Estimation Fused with Wearable Inertial Sensors. In *2020 15th IEEE International Conference on Automatic Face and Gesture Recognition (FG 2020)*. 508–514. <https://doi.org/10.1109/FG47880.2020.00092>
- Yuki Uno, Issei Ogasawara, Shoji Konda, Natsuki Yoshida, Naoki Otsuka, Yuya Kikukawa, Akira Tsujii, and Ken Nakata. 2023. Validity of Spatio-Temporal Gait Parameters in Healthy Young Adults Using a Motion-Sensor-Based Gait Analysis System (ORPHE ANALYTICS) during Walking and Running. *Sensors* 23, 1 (2023). <https://doi.org/10.3390/s23010331>